TSA6213 SYSTEM ADMINISTRATION & MAINTENANCE

LAB 7

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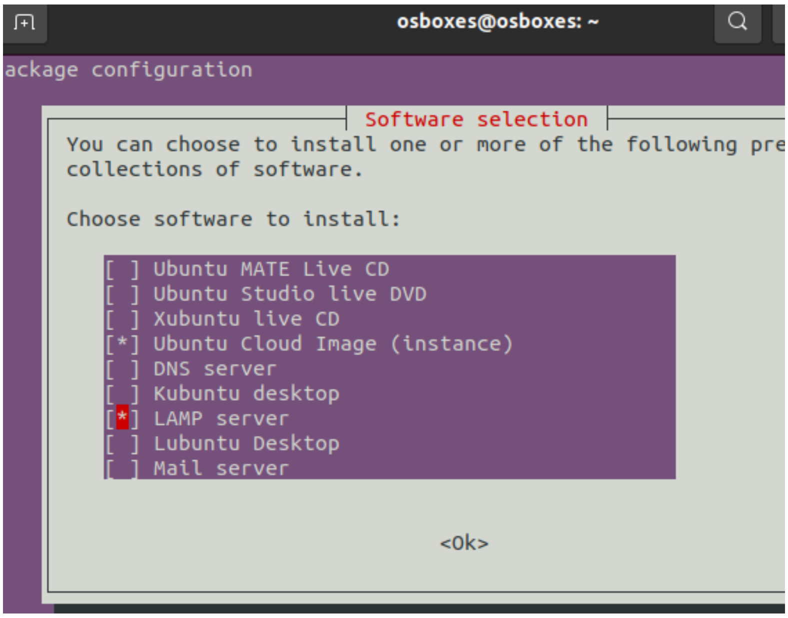
Apache and LAMP: Installation and Testing

1. Run your Linux virtual machine in VirtualBox and install LAMP server in your Linux machine using `sudo apt-get install lamp-server^` or `sudo tasksel` choose `LAMP server`. Install `tasksel` if command not found.

> sudo tasksel

> Find LAMP server, press spacebar to select. Press Enter.



```
osboxes@osboxes:~$ sudo apt-get install lamp-server^
```

```
osboxes@osboxes:~$ sudo apt-get install lamp-server^
Reading package lists... Done
Building dependency tree
Reading state information... Done
Note, selecting 'libgssapi3-heimdal' for task 'lamp-server'
Note, selecting 'libhttp-message-perl' for task 'lamp-server'
Note, selecting 'php7.4-cli' for task 'lamp-server'
Note, selecting 'libnghttp2-14' for task 'lamp-server'
Note, selecting 'libencode-locale-perl' for task 'lamp-server'
Note, selecting 'libwind0-heimdal' for task 'lamp-server'
Note, selecting 'mecab-utils' for task 'lamp-server'
Note, selecting 'libsasl2-modules-db' for task 'lamp-server'
Note, selecting 'libcurl4' for task 'lamp-server'
Note, selecting 'libldap-2.4-2' for task 'lamp-server'
Note, selecting 'libapache2-mod-php' for task 'lamp-server'
Note, selecting 'libssh-4' for task 'lamp-server'
Note, selecting 'libevent-core-2.1-7' for task 'lamp-server'
Note, selecting 'php-common' for task 'lamp-server'
Note, selecting 'libaprutil1' for task 'lamp-server'
Note, selecting 'libbrotli1' for task 'lamp-server'
Note, selecting 'php7.4-json' for task 'lamp-server'
Note, selecting 'mysql-client-8.0' for task 'lamp-server'
Note, selecting 'libheimntlm0-heimdal' for task 'lamp-server'
Note, selecting 'mysql-server' for task 'lamp-server'
Note, selecting 'mysql-server-8.0' for task 'lamp-server'
```

a) Test your Apache Installation. Open Apache server testing page in your web browser. Put current IP address or localhost/127.0.0.1 at your web browser to open the page.

> sudo systemctl status apache2

```
osboxes@osboxes:~$ sudo systemctl status apache2
Unit apache2.service could not be found.
```

> sudo apt install apache2 [if not installed]

```
osboxes@osboxes:~$ sudo apt install apache2
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libssh2-1 mc-data
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.2-0
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap
  liblua5.2-0
0 upgraded, 9 newly installed, 0 to remove and 156 not upgraded.
1 not fully installed or removed.
Need to get 1,829 kB of archives.
After this operation, 7,976 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 libapr1 amd64 1.6.5-1ubuntu1.1 [91.5 kB]
```

> sudo systemctl status apache2

```
osboxes@osboxes:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2024-12-26 02:32:19 UTC; 44s ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 17022 (apache2)
      Tasks: 55 (limit: 4592)
     Memory: 5.3M
     CGroup: /system.slice/apache2.service
             ├─17022 /usr/sbin/apache2 -k start
             ├─17024 /usr/sbin/apache2 -k start
             └─17025 /usr/sbin/apache2 -k start

Dec 26 02:32:19 osboxes systemd[1]: Starting The Apache HTTP Server...
Dec 26 02:32:19 osboxes apachectl[17018]: AH00558: apache2: Could not reliably determine the server's fully qualified
Dec 26 02:32:19 osboxes systemd[1]: Started The Apache HTTP Server.
```

> sudo systemctl enable apache2 [enable apache to start on boot]

```
osboxes@osboxes:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable apache2
```

> open localhost or 127.0.0.1 on web browser

b) Test the PHP installation.

• Write php test code `<?php phpinfo(); ?>` and give the file name as `info.php`

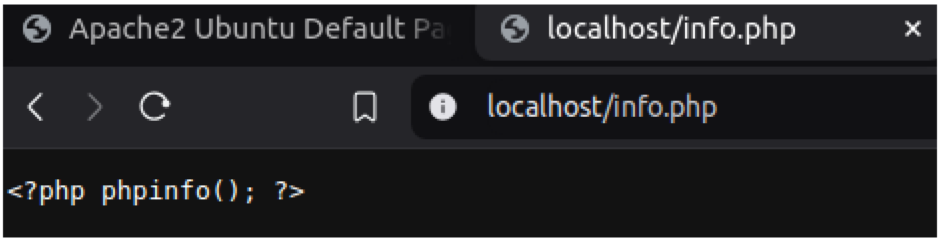
• View the code output in web browser in your Linux machine.

> sudo nano /var/www/html/info.php

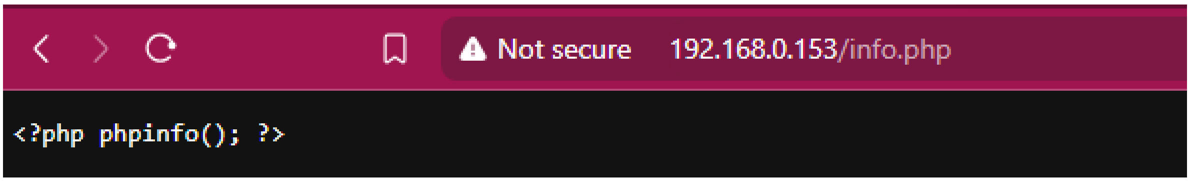
```
osboxes@osboxes:~$ sudo nano /var/www/html/info.php
```

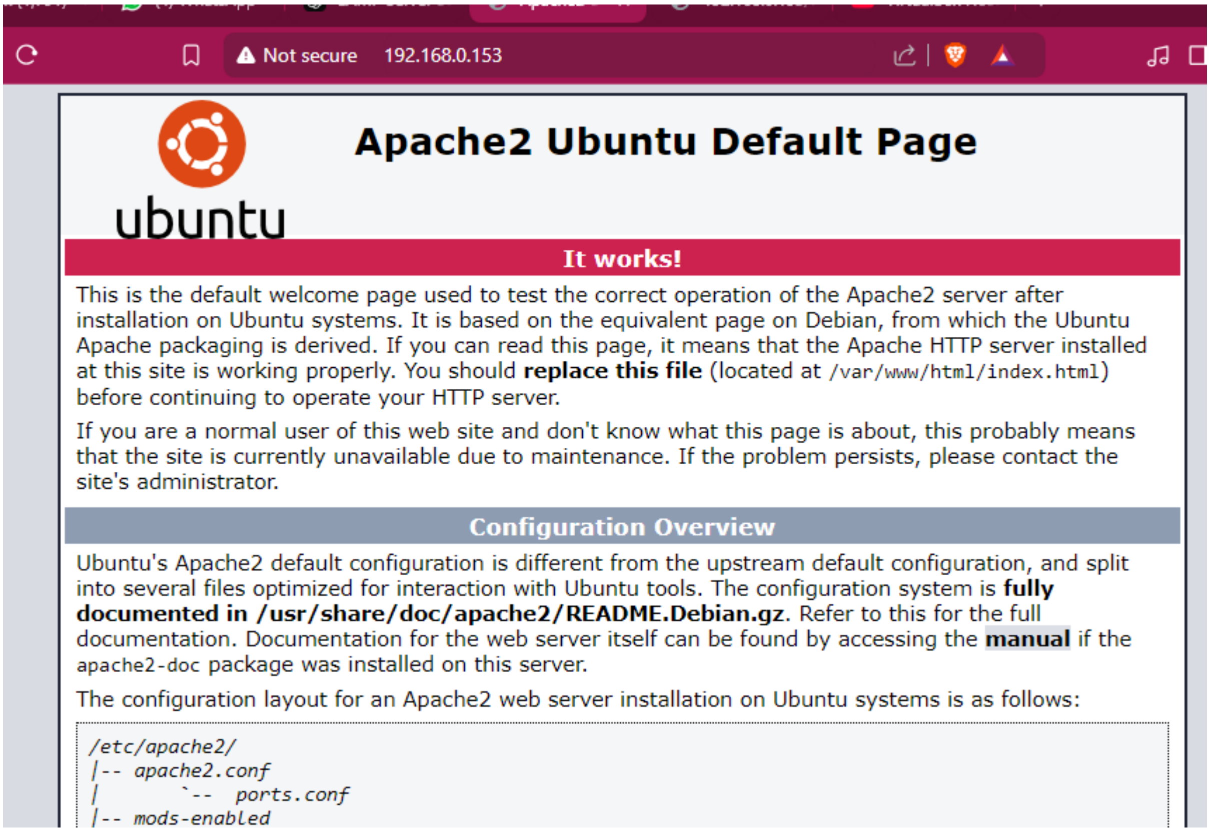
> add the code <?php phpinfo(); ?>

> save and close the file (Ctrl + X, Y, Enter).
> open the browser and open localhost/info.php



2. Set your Virtual Machine network to **Bridge or Host-only Adapter**. In windows host, test the Apache and PHP installation as in Q1a and Q1b using Bridge or Host-only Adapter IP address.





3. Create simple HTML page with body content as below:

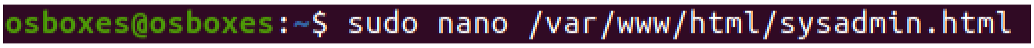
• **TSA6213** – header1 – h1, color blue

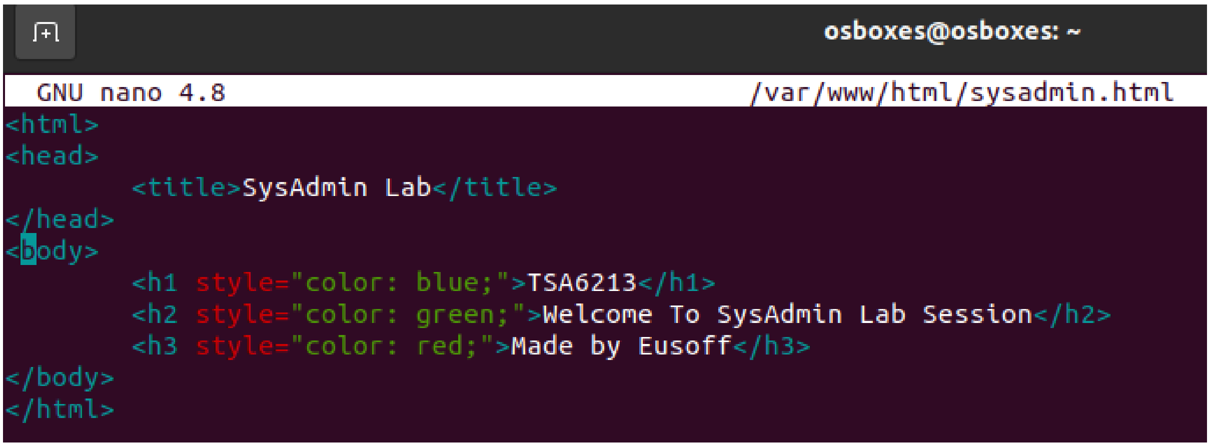
• **Welcome to SysAdmin Lab Session** as header 2 – h2, color green

• Save the files as `sysadmin.html` at `/var/www/html`.

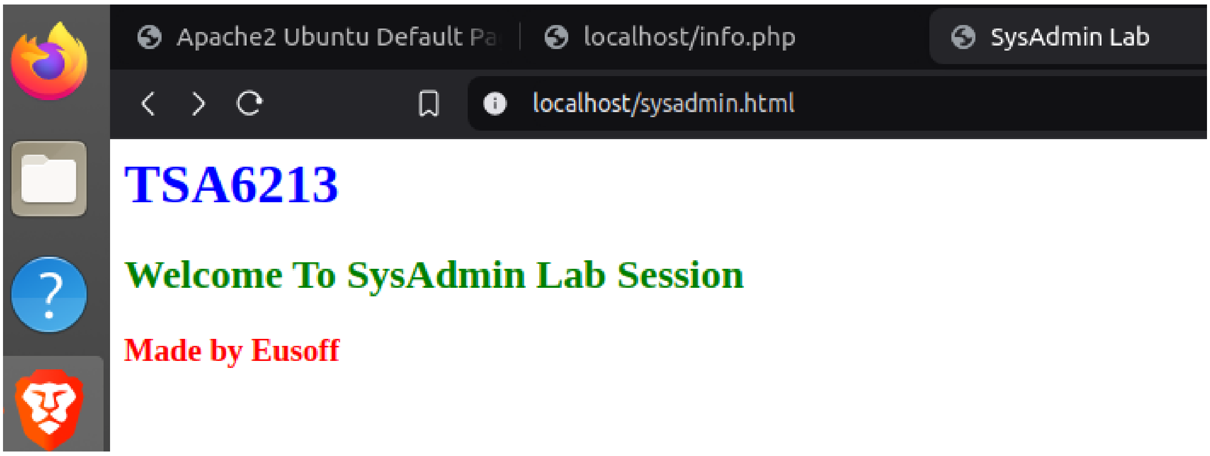
• View the file using Web Browser in Windows host.

> sudo nano /var/www/html/sysadmin.html

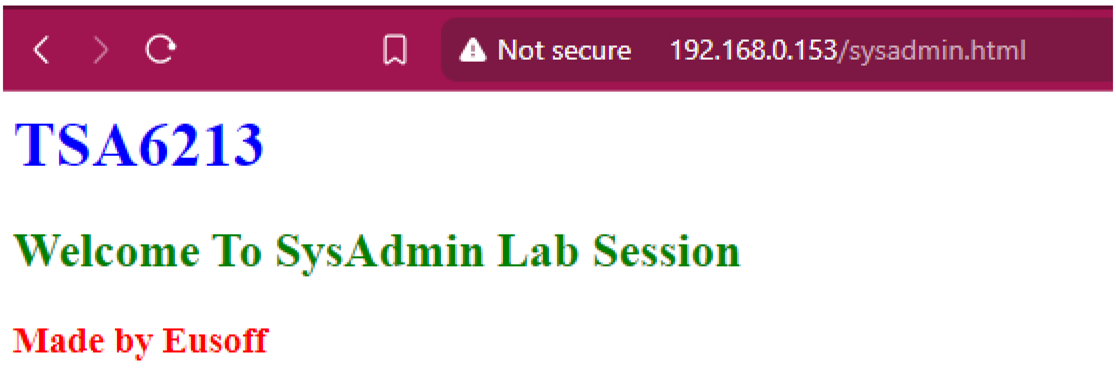




> From Ubuntu



> From Windows



4. Wiki is basically a website or database developed collaboratively by a community of users, allowing any user to add and edit content. The most popular wiki is Wikipedia, it uses mediawiki system. Many wiki system available for free and you can setup that easily in your web server, including mediawiki, dokuwiki, pmwiki and others. Setup and configure pmwiki system in your Apache web server.

a) Download the pmwiki from https://www.pmwiki.org/pub/pmwiki/pmwiki-latest.zip to your /var/www/html folder.
> go to directory cd /var/www/html
> sudo wget https://www.pmwiki.org/pub/pmwiki/pmwiki-latest.zip

```
eusoff@vbox:~$ cd /var/www/html
```

```
eusoff@vbox:/var/www/html$ sudo wget https://www.pmwiki.org/pub/pmwiki/pmwiki-latest.zip
--2025-01-19 16:28:38--  https://www.pmwiki.org/pub/pmwiki/pmwiki-latest.zip
Resolving www.pmwiki.org (www.pmwiki.org)... 23.254.203.248
Connecting to www.pmwiki.org (www.pmwiki.org)|23.254.203.248|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 671561 (656K) [application/zip]
Saving to: 'pmwiki-latest.zip'

pmwiki-latest.zip            100%[=====================================================>] 655.82K   489KB/s    in 1.3s

2025-01-19 16:28:40 (489 KB/s) - 'pmwiki-latest.zip' saved [671561/671561]
```

b) Install and setup the pmwiki with picture upload and admin password capability.
> sudo unzip pmwiki-latest.zip

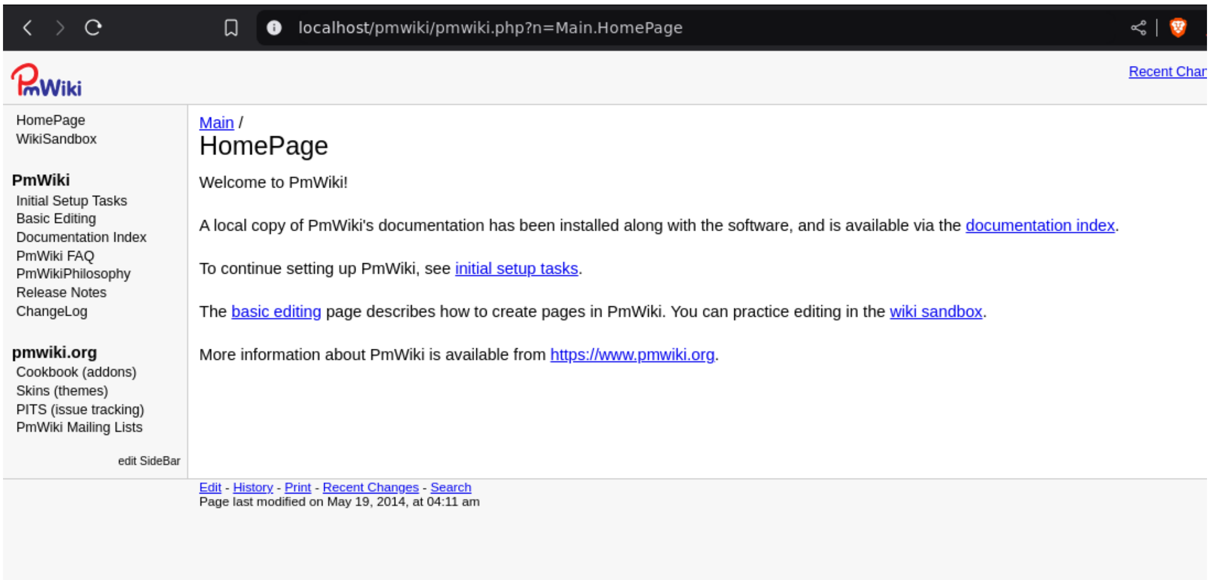
```
eusoff@vbox:/var/www/html$ sudo unzip pmwiki-latest.zip
Archive:  pmwiki-latest.zip
   creating: pmwiki-2.3.38/
   creating: pmwiki-2.3.38/cookbook/
  inflating: pmwiki-2.3.38/cookbook/.htaccess
  inflating: pmwiki-2.3.38/pmwiki.php
   creating: pmwiki-2.3.38/pub/
   creating: pmwiki-2.3.38/pub/guiedit/
  inflating: pmwiki-2.3.38/pub/guiedit/blank.gif
```

> ls
>sudo mv pmwiki-latest.zip pmwiki
>ls

```
eusoff@vbox:/var/www/html$ ls
dokuwiki  index.html  pmwiki-2.3.38  pmwiki-latest.zip  wordpress
eusoff@vbox:/var/www/html$ sudo mv pmwiki-2.3.38 pmwiki
eusoff@vbox:/var/www/html$ ls
dokuwiki  index.html  pmwiki  pmwiki-latest.zip  wordpress
```

> sudo chown -R www-data:www-data /var/www/html/pmwiki
> sudo chmod -R 755 /var/www/html/pmwiki

```
eusoff@vbox:/var/www/html$ sudo chown -R www-data:www-data /var/www/html/pmwiki
eusoff@vbox:/var/www/html$ sudo chmod -R 755 /var/www/html/pmwiki
```



c) Edit the default pmwiki page and replace it with:

- Title Page/Header: “System Administration and Maintenance”. **Text color is blue**.
- Text: “**Lab** Practical”. **Lab is bold and Practical is underlined.**
- Create bullet list with three items: **Unix, Linux, FreeBSD**
- Create number with four items: **Ubuntu, Kubuntu, Lubuntu, Xubuntu**
- Create a link to fist.mmu.edu.my
- Image/ MMU Logo. Upload the image into the server and display the logo in wiki.

localhost/pmwiki/pmwiki.php?n=Main.HomePage

PmWiki

HomePage
WikiSandbox

PmWiki

Initial Setup Tasks
Basic Editing
Documentation Index
PmWiki FAQ
PmWikiPhilosophy
Release Notes
ChangeLog

pmwiki.org

Cookbook (addons)
Skins (themes)
PITS (issue tracking)
PmWiki Mailing Lists

edit SideBar

Main /

System Administration and Maintenance

System Administration and Maintenance

Lab Practical

- Unix
- Linux
- FreeBSD

1. Ubuntu
2. Kubuntu
3. Lubuntu
4. Xubuntu

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